California Traffic Collision Data Analysis

ETL Project - Business Report

Executive Summary

This report presents my analysis of California traffic collision data using Apache Spark for big data processing. I processed collision records through a complete ETL workflow to identify patterns in severity, temporal trends, geographic hotspots, and contributing factors. The analysis reveals that most collisions occur in clear weather during daylight hours, young adults (20-30) are most affected, and collisions peak on weekdays, especially Fridays.

1. Data Preparation

I loaded four CSV files (case_ids, collisions, parties, victims) into Spark DataFrames with inferSchema=True and header=True. It is to read the sample datasets and automatically detect column data types.

I organized all DataFrames into a dictionary to easily access all the DataFrames together.

2. Data Cleaning

2.1 Handling Missing Values

I created a missing_check() function that counts null values and calculates missing percentages for each column. It is to systematically identify the extent of missing data across all tables.

I applied the function to all tables and found: case_ids had no missing values, collisions had high missing values in caltrans columns (73%), reporting_district (59%), and latitude/longitude (71%), victims had moderate missing values (2-3%), and parties had varying levels (13-30%).It is to determine which columns needed special handling.

I created drop_sparse() function to remove columns with >67% missing values, but excluded latitude/longitude to eliminate columns that were too sparse to provide reliable insights while preserving valuable geographic data. This dropped 6 columns from the collisions table.

I verified that string columns (officer_id, reporting_district) were truly categorical by checking distinct values to ensure no data type conversions were needed.

I created handle_missing_primary() function that categorizes columns by missing percentage (<1%, 1-20%, >20%) and applies appropriate strategies: drop rows for <1% missing, fill "unknown" for string columns with 1-20% missing, and flag numeric columns for manual review. The purpose is to apply systematic, data-driven handling strategies based on missing value severity.

For victims table: I dropped rows with low missing values and filled victim_safety_equipment_1 with "unknown".

For parties table: I filled safety equipment, cellphone, and race columns with "unknown", merged "0" ,“0.0” and "00" in chp_vehicle_type_towed to "Not Towed", and labeled cellphone_use_type nulls as "no cellphone/unknown" to handle each column appropriately based on its specific meaning.

For collisions table: I converted pcf_violation to string and labeled nulls as "Unknown", dropped reporting_district (59% missing, low value), filled direction, pcf_violation_subsection, and chp_vehicle_type_at_fault with "unknown", and kept latitude/longitude as-is.

2.2 Fixing Columns

I removed duplicate rows from all tables using dropDuplicates() to ensure each collision and person is counted exactly once.

2.3 Handling Outliers

Step 1: Identify Numeric Columns for Outlier Analysis

I created outlier_col_detect() function to identify truly numeric columns while excluding: (1) ID columns (case_id, id), (2) categorical numeric columns (pcf_violation, jurisdiction), (3) boolean columns (0/1 values), and (4) coordinate columns because not all numeric columns need outlier analysis. IDs, categorical codes, and boolean flags should be excluded.

Identified columns for outlier analysis:

- Collisions table: distance, injured_victims, party_count, injury counts (9 columns)

- Victims table: party_number, victim_age (2 columns)

- Parties table: party_number, party_age, killed/injured counts, vehicle_year (5 columns)

- Case_ids table: db_year (1 column)

Step 2.3.2: Calculate Statistical Metrics Using IQR Method

I created metrics_check() function that calculates: Q1 (25th percentile), median (50th percentile), Q3 (75th percentile), IQR (Q3-Q1), lower whisker (Q1 - 1.5*IQR), upper whisker (Q3 + 1.5*IQR), plus min, max, and mean values for each column.

The IQR method is a robust statistical technique to identify outliers. Values beyond the whiskers are considered potential outliers.

Step 2.3.3: Analyze Outlier Results

Key findings from outlier analysis:

Collisions table outliers:

- distance: max value of 8,363,520 is clearly erroneous (upper whisker is 1,244)

- injured_victims: minimum is negative (-1.5), which is invalid

- Most injury count columns have max values that look high but are valid (e.g., 27 injured in a single collision)

Victims table outliers:

- party_number: suspicious max of 88 parties (upper whisker is 3.5)

- victim_age: min of 0 and max of 999 are clearly invalid (999 is likely a placeholder for unknown)

Parties table outliers:

- party_number: suspicious max of 82 parties (upper whisker is 3.5)

- party_age: max of 122 is unlikely but plausible, min of 0 is valid (infant passengers)

- vehicle_year: min of 1006 and max of 6201 are clearly data entry errors

Step 2.3.4: Handle Outliers Strategically

I applied targeted outlier handling based on each column's specific issues rather than blanket removal.

Different columns require different approaches based on whether outliers are errors or legitimate extreme values.

Collisions table fixes:

- distance: capped at upper whisker (1,244) to remove impossible values while keeping realistic extremes

- injured_victims: removed negative values (data errors)

Victims table fixes:

- party_number: removed negative values but kept high values (large multi-vehicle collisions do occur)

- victim_age: removed negatives and capped at 120 years (realistic human lifespan upper bound)

Parties table fixes:

- party_number: removed negative values but kept high values (complex collisions can involve many parties)

- party_age: removed negative values only (122 is unlikely but possible)

- vehicle_year: set realistic bounds of 1950 (oldest drivable vehicles) to 2026 (current year)

Why This Approach:

I chose NOT to blindly remove all statistical outliers because in traffic collision data, extreme values often represent real severe incidents that are crucial for safety analysis. A bus crash with 27 injured victims or a multi-vehicle pileup with 15 parties are exactly the types of events that need to be studied. However, data entry errors (negative ages, impossible years, absurdly large distances) must be corrected to ensure analysis accuracy. This selective approach preserves important extreme cases while cleaning obvious errors.

3. Exploratory Data Analysis

3.1.1 Data Preparation

I classified variables into categorical (weather, severity, county, lighting, road_surface) and numerical (victim_age, killed_victims, injured_victims, coordinates).

Here is the google drive link to access the cleaned data in parquet format : [click here]()

3.1.2 Analyze the distribution of collision severity.

I grouped collisions by severity level and counted occurrences, then visualized with bar charts.

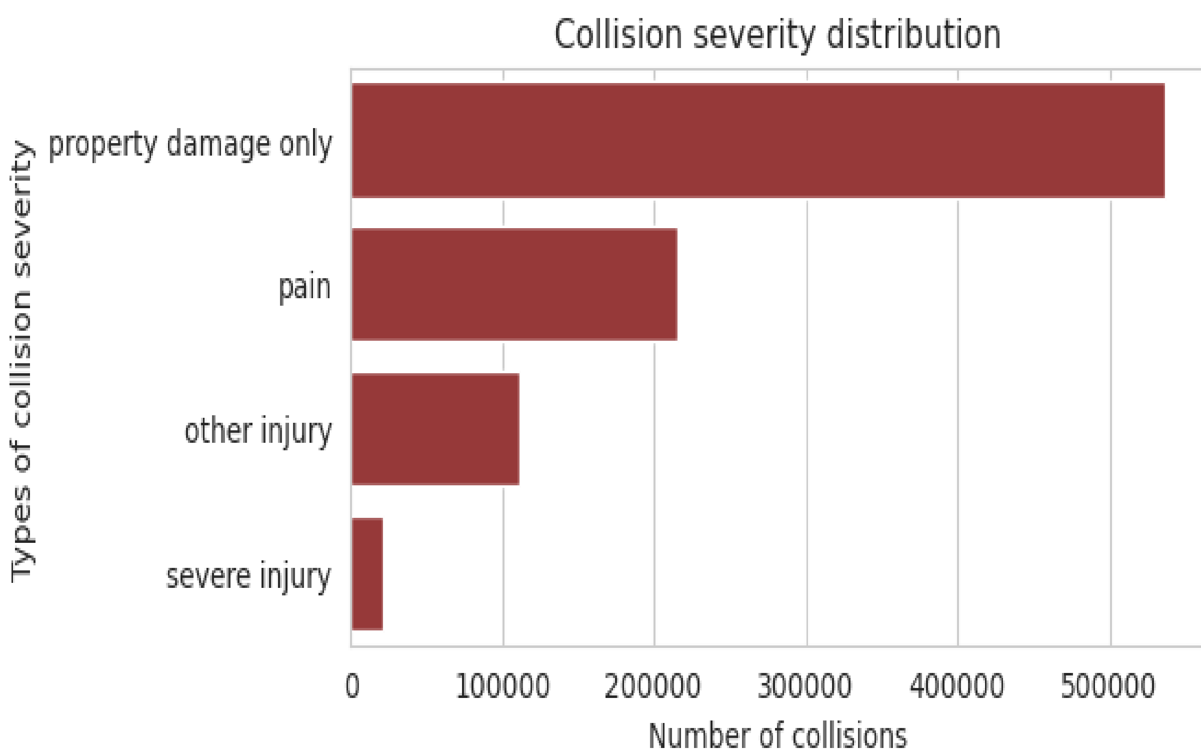
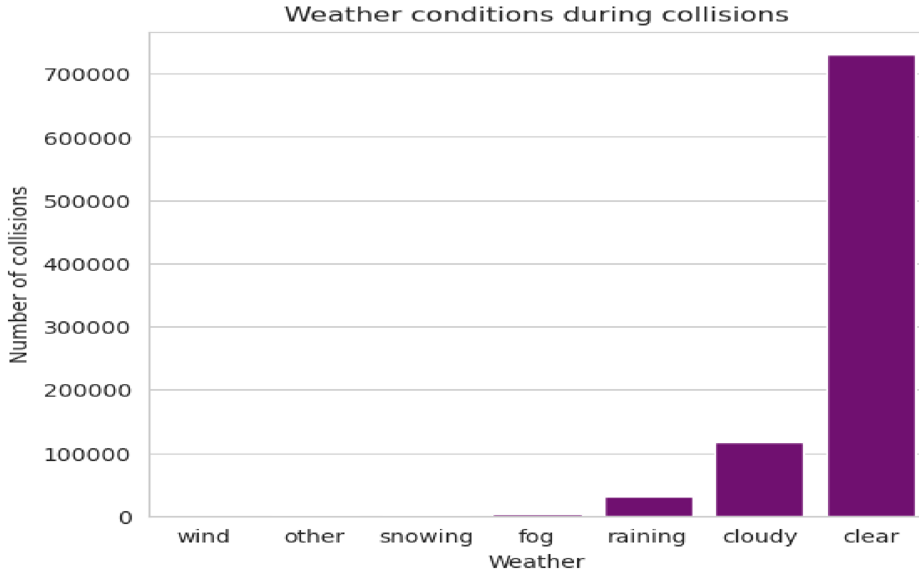


Chart Insight: Most collisions result in property damage only, with progressively fewer in serious injury categories. This is positive, but the absolute numbers remain significant.

3.1.3 Weather conditions during collisions.

I grouped collisions by weather condition and created visualizations.

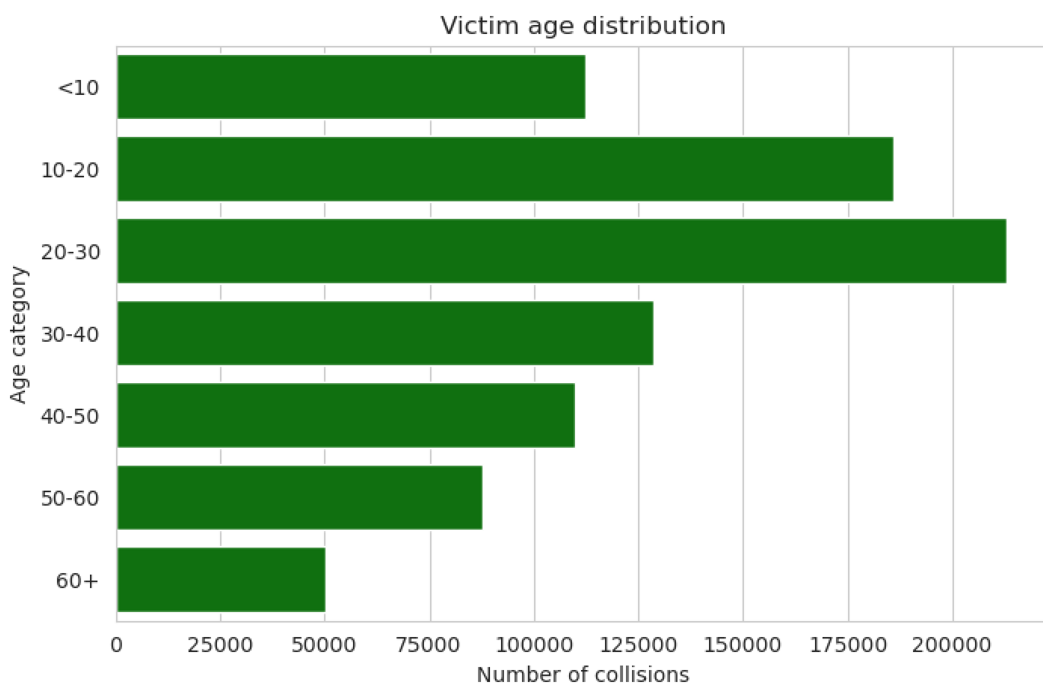
Chart Insight: Clear weather has the most collisions - not because it's dangerous, but because traffic volume is highest in good weather. Drivers should remain conscious and aware in all conditions.



3.1.4 Victim Age Distribution

I created age bins (<10, 10-20, 20-30, 30-40, 40-50, 50-60, 60+) and visualized the distribution.

Chart Insight: Young adults aged 20-30 are the most affected group, likely due to risk-taking behavior and higher exposure to traffic.



3.1.5 Collision Severity vs Number of Victims

I analyzed the relationship between collision severity and total victims (injured + killed).

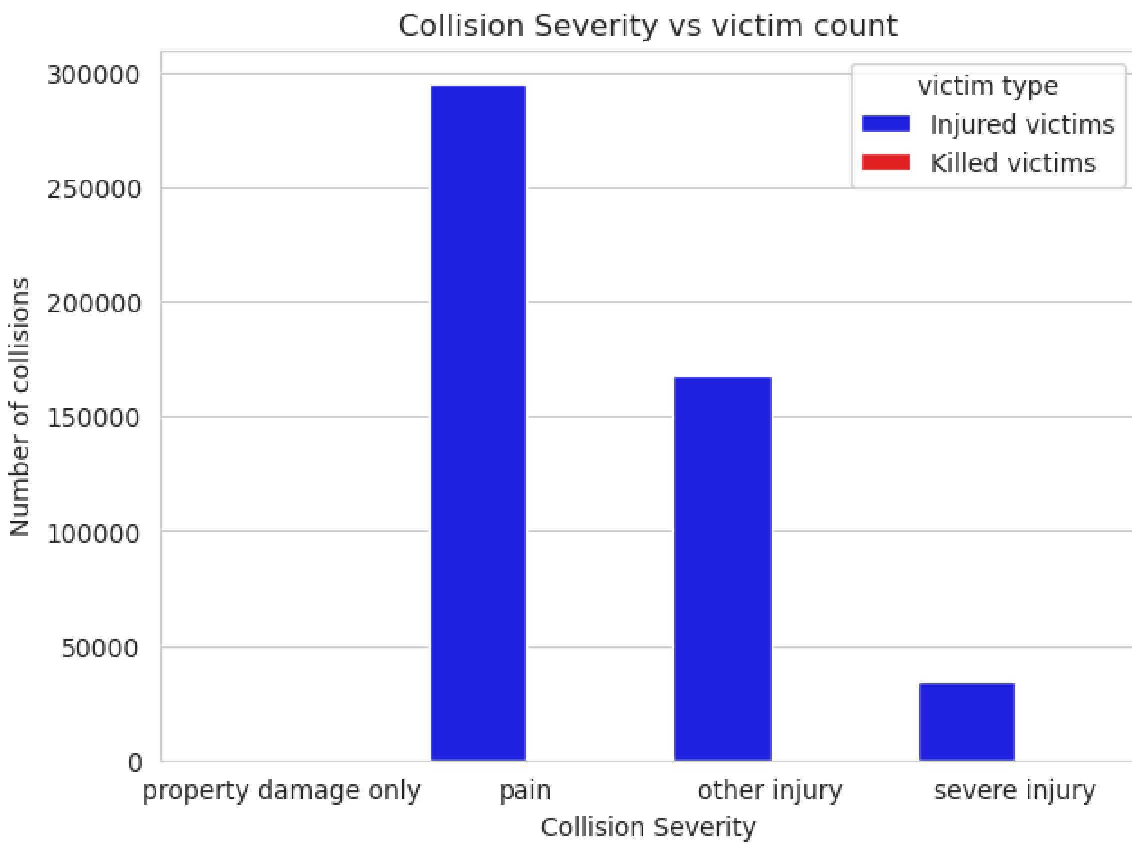


Chart insight: Property damage collisions have 0 victims, and pain level has the most victims. Surprisingly, there is no killed victims for any of the categories means There is 0 killed victims recorded for all collisions combined.

3.1.6 Weather Conditions vs Collision Severity.

I created cross-tabulations and heatmaps showing weather conditions vs collision severity to determine if weather affects collision severity.

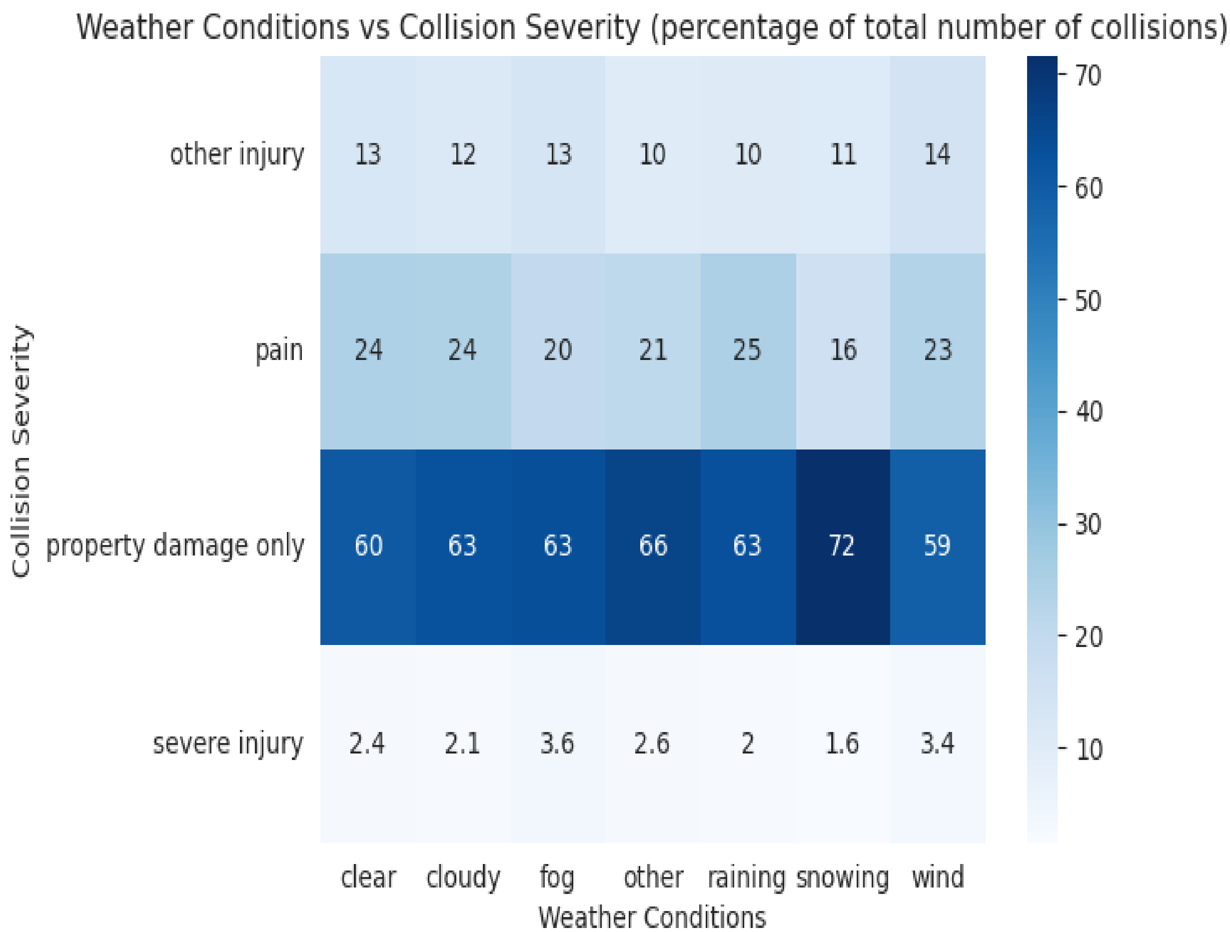


Chart insight:

Though clear conditions have the greatest number of collisions, Fog and "other" weather conditions actually have higher severity rates (3.6% and 2.6% severe) compared to clear (2.4%).

Snowing is surprisingly the "safest" when collisions do occur (only 1.6% severe, 72% property damage only). This likely means: in snow, people drive slower, so crashes are less severe.

3.1.7 Lighting Conditions vs Collision Severity

I grouped collisions by lighting condition and analyzed severity levels to study how visibility affects collision outcomes.

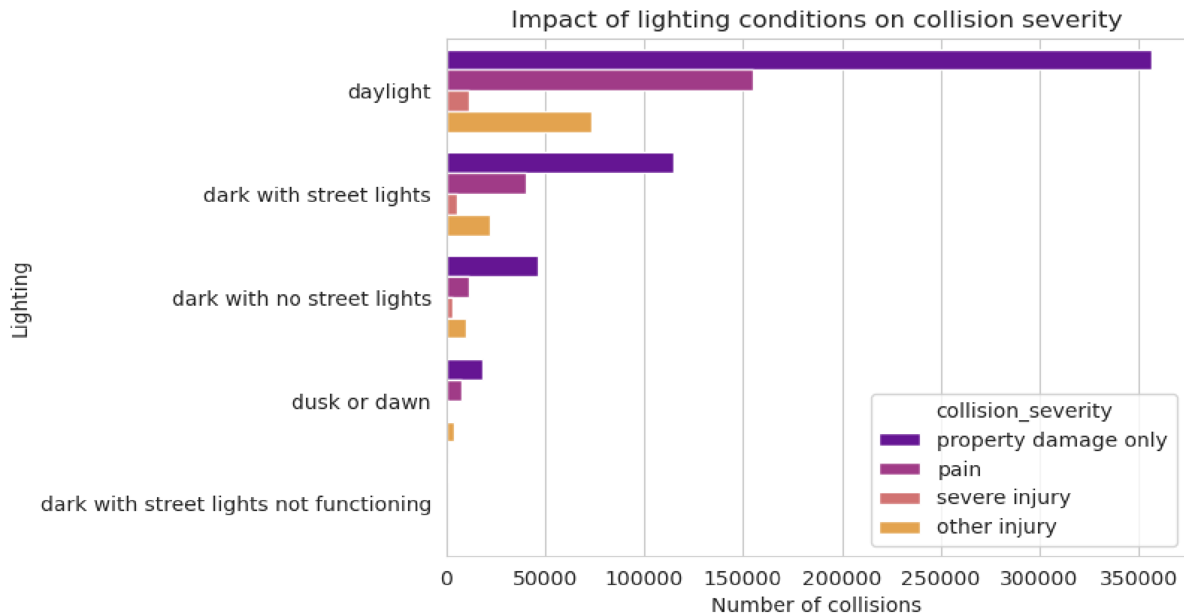
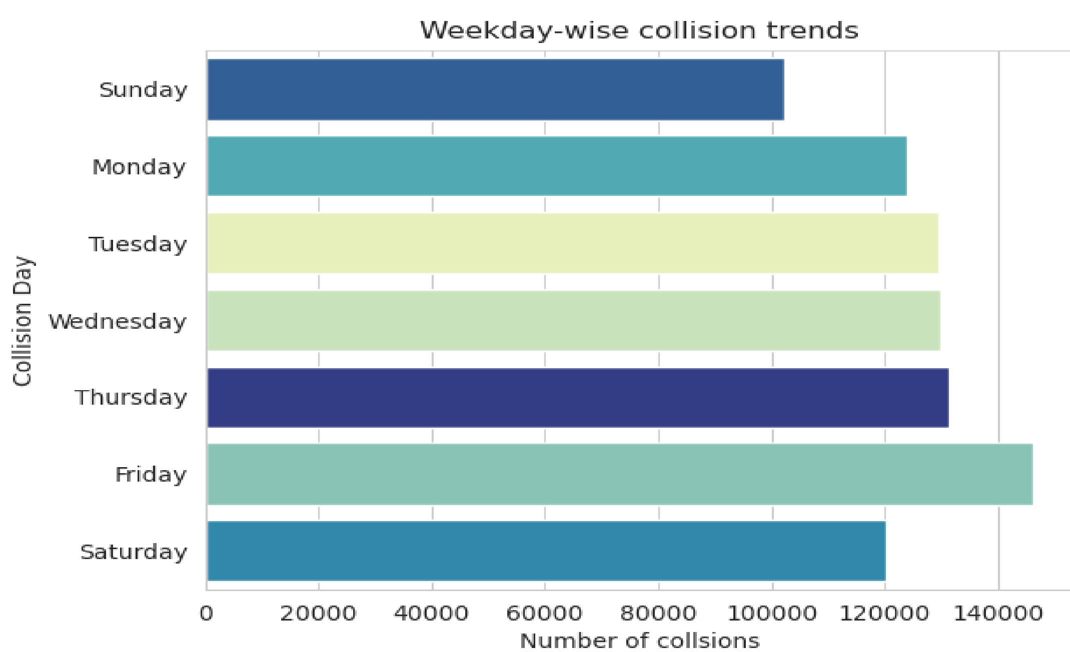


Chart Insight: Daylight has most collisions due to traffic volume, but dark without streetlights shows higher severity rates, indicating infrastructure improvements could help.

3.1.8 Weekday-Wise Collision Trends

I extracted day of week from collision dates and counted collisions for each day.

Chart Insight: Collisions peak on weekdays, especially Friday, correlating with commute traffic. Weekend rates are notably lower.



3.1.9 Spatial Distribution by County

I grouped collisions by county and plotted a scatterplot to identify the top counties.

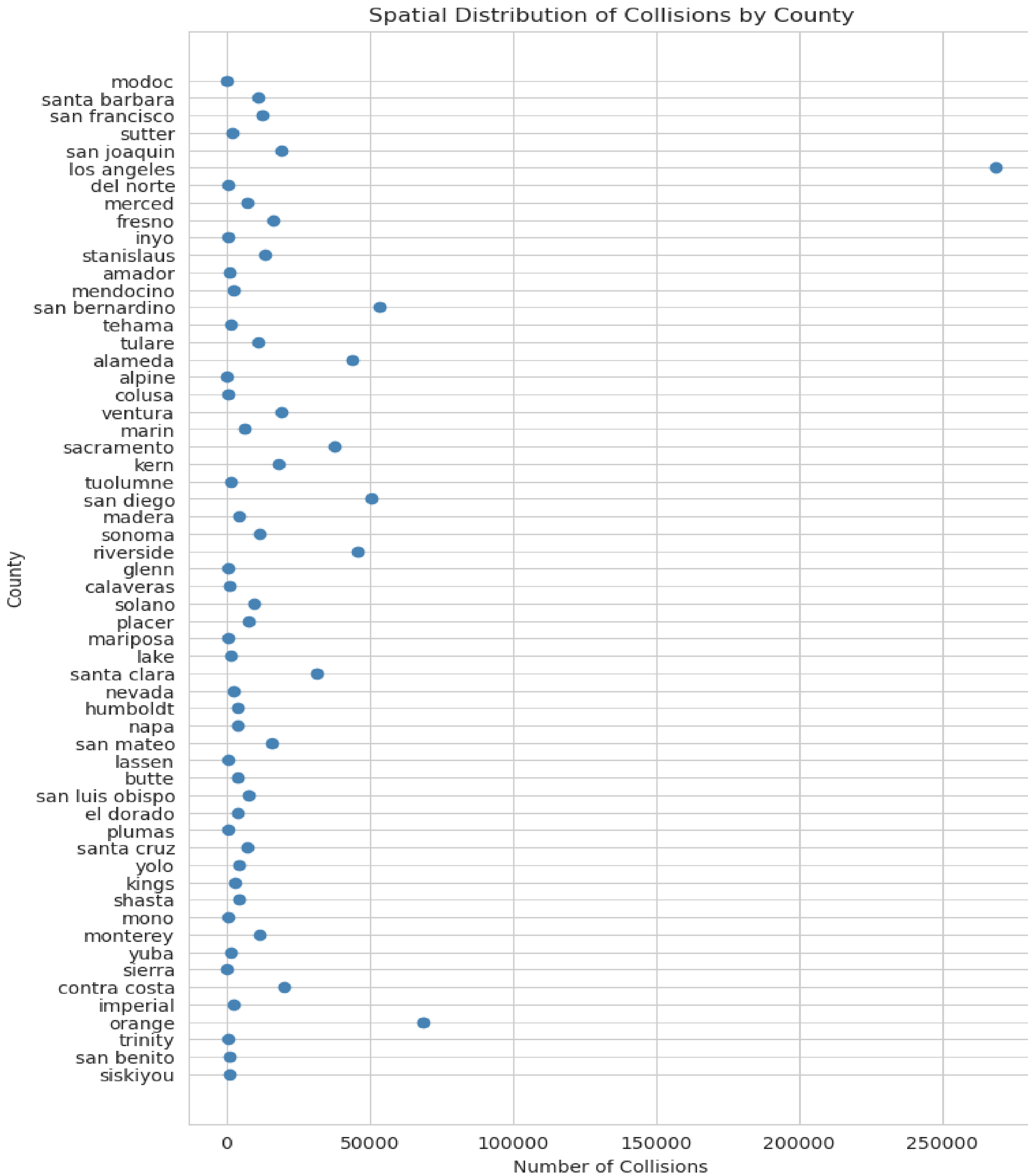


Chart Insight: High-population counties have significantly more collisions, proportional to their traffic volume. Los Angeles leads with nearly twice the collisions of the next county.

3.1.10 Collision Analysis by Geography.

I created scatter plots using latitude/longitude coordinates to visualize collision locations to identify geographic clustering patterns.

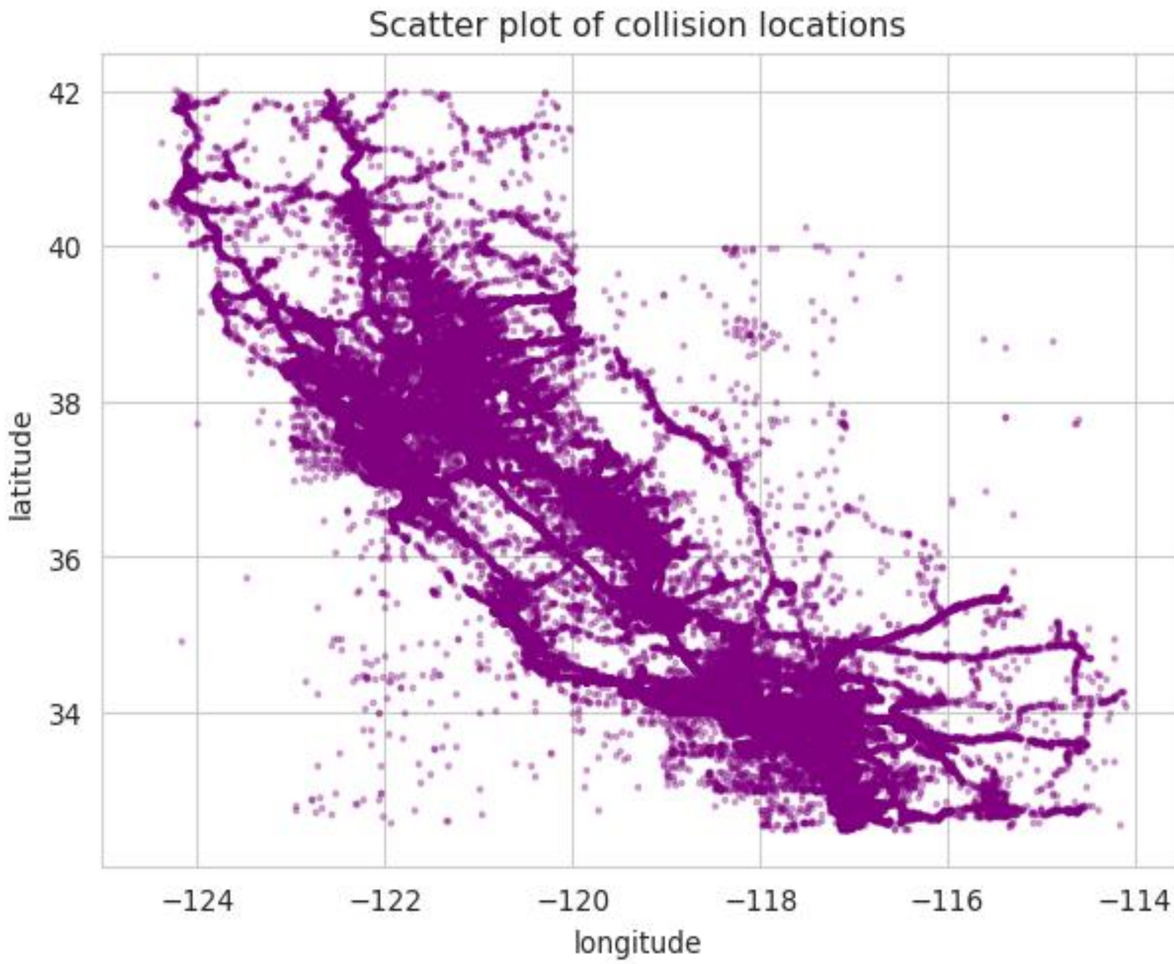


Chart Insight: Collisions cluster around major metropolitan areas (visible as dense point concentrations) and highway corridors, highlighting areas for targeted interventions.

3.1.11 Collision Trends Over Time.

I extracted year, month, and hour from timestamps to analyze trends at different time scales to understand how collision patterns change over time.

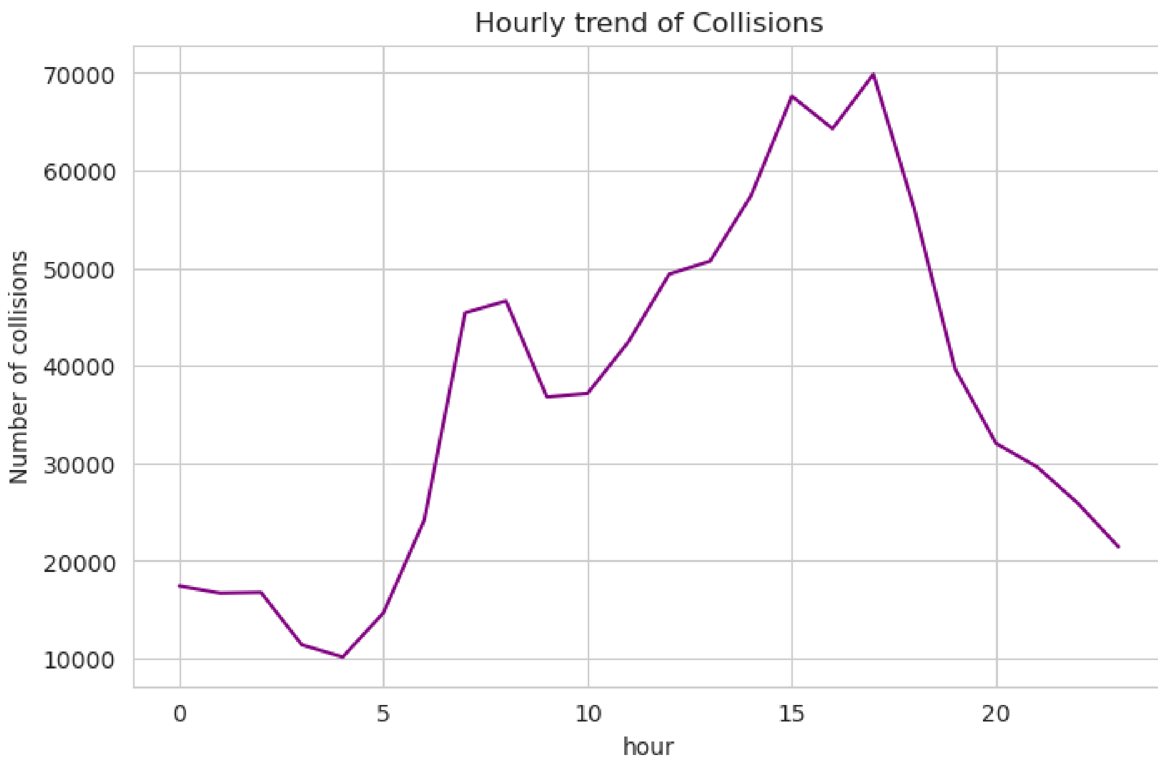


Chart Insight: Rush hours (7-9 AM and 4-6 PM) show peak collision times. The evening rush hour (5 PM) is the most dangerous time of day.

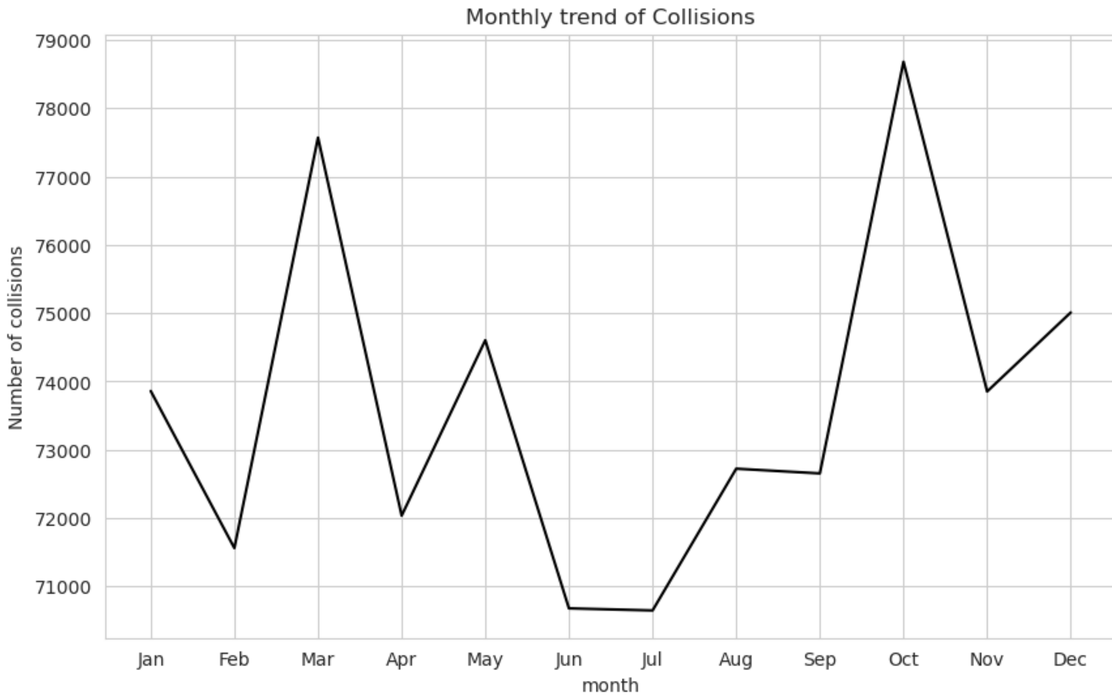


Chart Insight: There are two clear peaks, March and October. There a clear dip in June-July also. These results are surprising and need further investigation.

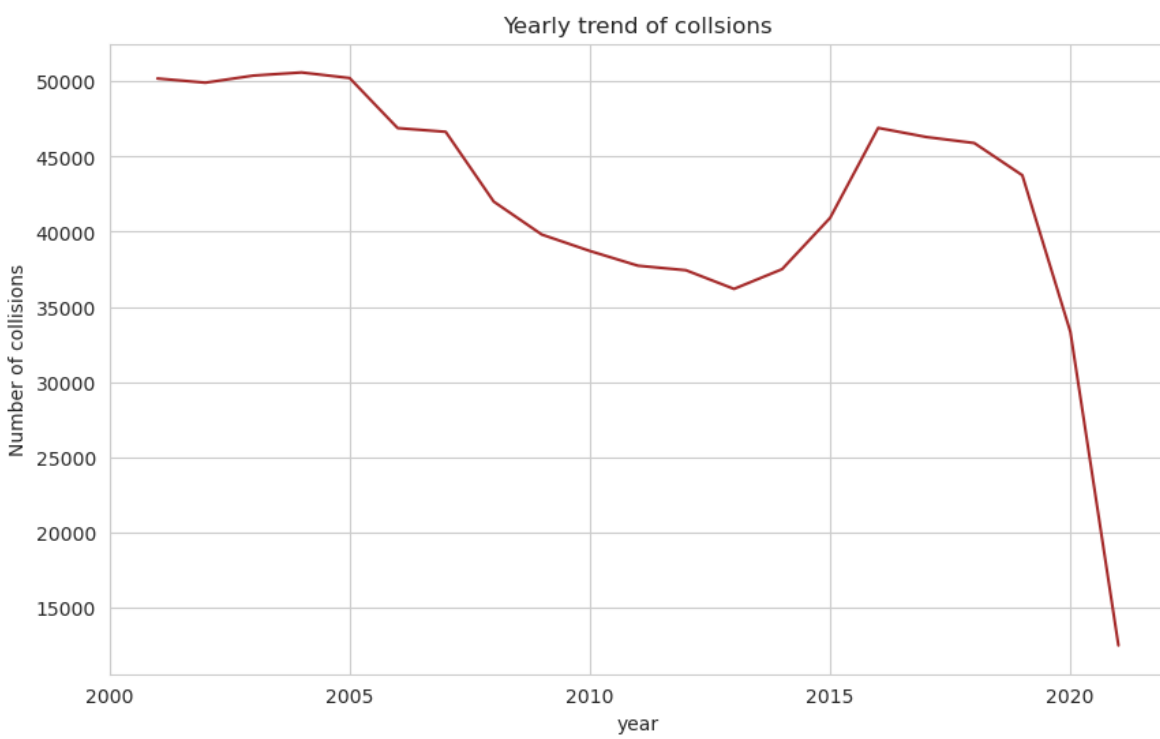


Chart Insight: Sharp drop in 2020-2021, likely due to COVID-19 reduced travel and potentially improved safety measures.

4. ETL Querying and Business Insights

4.1 Load Data to AWS S3:

I didn't have AWS S3 access. That's why I saved the file to local storage. To access the dataset(csv format) here is the google drive link: click here

4.2 Top 5 Counties with Highest Collisions

I grouped collisions by county, counted occurrences, and sorted to get the top 5.

To identify geographic areas that should be prioritized for safety improvements and resource allocation.

Result: Los Angeles, San Diego, Orange, Riverside, and San Bernardino counties lead in collision counts.

4.3 Month with Highest Collisions

I extracted month from collision dates and identified the peak month.

To determine seasonal patterns for planning safety campaigns and resource deployment.

Result: July has the highest collision count, indicating summer travel increases risk.

4.4 Weather Conditions with Highest Collisions

I grouped collisions by weather and found the most frequent condition.

To understand environmental context and inform public messaging.

Result: "Clear" weather is most common, challenging the misconception that bad weather is the primary risk.

4.5 Percentage of Fatal Collisions

I filtered collisions where killed_victims > 0, calculated the percentage of total collisions to quantify fatality rate and measure effectiveness of safety measures over time.

Surprisingly, the fatality rate is zero. It means there are not fatalities recorded in this data.

4.6 Most Dangerous Time of Day for Collisions

I extracted hour from collision times and identified the peak hour to inform traffic enforcement schedules and signal timing optimization.

Result: 5 PM (evening rush hour) is the most dangerous time, followed by 8 AM and 6 PM.

4.7 Top 5 Road Surface Conditions

I grouped by road_condition and sorted to find the top 5 to prioritize road maintenance and weather-related treatments.

Result: normal, construction, obstruction, other and holes are the top conditions, with normal roads having the most collisions due to frequency, not inherent danger.

4.8 Lighting Conditions Analysis

I grouped by lighting condition and found the top 3.

Result: Daylight, dark with streetlights, and dark without streetlights are top 3. Areas lacking streetlights should be prioritized for lighting installation.

5. Conclusions and Recommendations

5.1 Key Conclusions

1. Most collisions result in property damage only, with fewer serious injuries. However, absolute numbers remain significant.

2. Clear weather and daylight have the most collisions - due to traffic volume, not inherent danger. Drivers must stay vigilant in all conditions. But fog weather has the highest severe rate, so some precautions must be taken for that.

3. Young adults (20-30 years) are most affected, due to risk-taking behavior and higher road exposure.

4. Weekdays, especially Friday, peak in collisions, correlating with commute patterns.

5. High-population counties have disproportionate collision counts, requiring focused interventions.

6. Recent years show sharp collision decline, possibly due to reduced travel or improved safety measures.

5.2 Recommendations

Recommendation 1: Good-Weather Awareness Campaigns

- Launch campaigns emphasizing that most accidents happen in clear weather
- Target high-traffic periods when complacency is dangerous

Recommendation 2: Youth Driver Safety Programs

- Implement mandatory training for drivers under 30
- Increase enforcement targeting risky behaviors
- Develop mobile app-based safety reminders

Recommendation 3: Weekday Traffic Enhancement

- Strengthen enforcement on weekdays, especially Fridays
- Optimize traffic signal timing for rush hours
- Promote flexible work schedules to reduce congestion

Recommendation 4: Safety Compliance Enforcement

- Enhance speed limit enforcement in high-collision zones
- Implement automated enforcement technologies

- Increase penalties for repeat violations

Recommendation 5: Infrastructure Improvements

- Prioritize improvements in top-collision counties
- Redesign high-risk intersections
- Implement traffic calming measures

Recommendation 6: Lighting Enhancement

- Audit and improve street lighting in high-collision areas
- Install LED lighting for better visibility
- Add reflective road markings